Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene

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The large superfluid response reported in magic-angle twisted bilayer graphene is often discussed in terms of quantum geometry, but it remains useful to ask whether the orbital structure of the superconducting gap leaves a separable finite-band response diagnostic. We answer this question with an executable Bistritzer-MacDonald–Bogoliubov-de Gennes workflow for orbital-projected interband-pairing directions. The pairing family is defined in the layer-sublattice basis and projected into the normal-state band basis using a fixed time-reversal-sewn valley convention, so that pairing direction, band projection, and response normalization can be varied separately. For the working direction $M_1 = \tau_x\sigma_x$, the projected interband pairing weight increases smoothly with the tuning parameter η and reaches approximately 0.108 at $\eta = 1$ for the $n_{\mathrm{keep}} = 6$ truncation. At fixed Frobenius norm of the orbital pairing matrix, dense scans over chemical potential and pairing interpolation give a normalized isotropic response change of approximately $+0.56\%$ to $+11.86\%$ between $\eta = 0$ and $\eta = 1$ over $\mu = -5, \ldots, 5$ meV; selected $nk = 9, 11, 13$ checks and a representative $nk = 15$ spot check preserve the positive fixed-Frobenius sign. The fixed raw-gap-amplitude control is much weaker and can be negative. We therefore identify the signal as a normalization-conditioned mechanism signature rather than an unconditional enhancement of the absolute stiffness. An absolute-unit audit of a previous PRB-style superfluid-weight benchmark shows that the legacy values can be only partially reconstructed from one unified convention. The resulting manuscript policy is that normalized response maps carry the main mechanism claim, while absolute stiffness comparisons are deferred to a separate benchmark calculation with a fully declared convention.

I. INTRODUCTION

Magic-angle twisted bilayer graphene (MATBG) provides a setting in which superconductivity, topology, flat bands, and quantum geometry meet in a single experimentally accessible platform [1–3]. In conventional superconductors the superfluid weight is tied to band dispersion and carrier density, but in flat or nearly flat bands a geometric contribution associated with the quantum metric can remain finite even when ordinary band velocity is small [4, 5]. This geometric contribution is especially relevant for MATBG, whose fragile topology and $C_{2z}\mathcal{T}$ symmetry protect nontrivial band geometry in the flat-band manifold [6–8].

Circuit-QED measurements have reported a superfluid stiffness in MATBG that is large compared with a simple BCS estimate [9]. More recent transport, microwave, and radio-frequency Josephson-junction measurements have directly accessed superfluid stiffness or condensate dynamics and point to nontrivial low-temperature gap structure [10, 11]. Related twisted-trilayer stiffness measurements reinforce the broader nodal-gap phenomenology in magic-angle graphene superconductors [12]. Several theoretical works have connected this anomaly to quantum geometry, remote-band effects, and beyond-BCS corrections [13–15]. Recent electron–phonon, dielectric-environment, molecular-pairing, and strong-correlation studies further emphasize that the microscopic pairing mechanism in twisted bilayer graphene is not yet settled [16–19]. In parallel, band-off-diagonal superconductivity has been proposed as a candidate state in twisted graphene superlattices, and its Eliashberg superfluid-stiffness signatures have recently been analyzed [20, 21]. Yet a practical question remains: if a mean-field pairing ansatz contains interband structure after projection into the MATBG band basis, does that structure leave a robust and separable signature in the computed superfluid response?

This paper answers that question with an explicitly executable finite-band workflow. Rather than starting from a band-basis off-diagonal gap chosen by hand, we define the pairing matrix in the layer-sublattice orbital basis and project it into the normal-state band basis. This keeps the pairing direction covariant under band-basis gauge changes and separates two issues that are often entangled: how interband pairing is defined, and how the response is normalized before it is interpreted. The main output is a normalized mechanism map,

$$\delta D_{\mathrm{iso}}(\mu, \eta) = \frac{D_{\mathrm{iso}}(\mu, \eta)}{D_{\mathrm{iso}}(\mu, 0)} - 1, \quad (1)$$

computed for two gap-normalization controls.

The central result is deliberately conservative. At fixed Frobenius norm of the orbital pairing direction, increasing the interband-pairing component produces a reproducible change in the isotropic response across chemical potential, with selected higher-grid checks supporting the sign and scale of the effect. At fixed raw gap amplitude, the effect is much smaller and sign-sensitive. Thus the present calculation supports a normalization-conditioned

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mechanism signature, not a universal statement that interband pairing always enhances the physical superfluid stiffness. We also perform a self-audit of the absolute units and a previous PRB-style benchmark table; that audit sets the observable policy used below. Normalized response ratios are the claim-bearing quantities in this mechanism paper, whereas absolute stiffness tables should be regenerated from a single declared convention before being used for final experimental comparison.

Figure 1 summarizes the executable route used throughout the paper. The diagram is a reproducibility map rather than an additional numerical result: it identifies where the pairing direction is defined, where band projection enters, where the finite-band response is evaluated, and which output is allowed to carry the mechanism claim.

II. MODEL AND PAIRING CONSTRUCTION

A. Bistritzer-MacDonald model

We use a single-valley Bistritzer-MacDonald continuum Hamiltonian with momenta in Å^{-1} and energies in meV [22]. The basis for each moire reciprocal vector $\mathbf{G}$ is

$$(tA, tB, bA, bB), \tag{2}$$

where t, b label top and bottom layers and A, B label sublattices. Unless otherwise stated, the numerical scans use

$$\begin{aligned} \theta &= 1.05^\circ, & v_F &= 2.135 \text{ eV Å}, \\ w_0 &= 87.2 \text{ meV}, & w_1 &= 109.0 \text{ meV}, \end{aligned} \tag{3}$$

with a square reciprocal cutoff $N_{\text{shell}} = 3$, giving a 196×196 normal-state Hamiltonian per valley and spin. The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \text{ Å}^2, \quad G_M = 0.054047 \text{ Å}^{-1}. \tag{4}$$

B. Orbital-defined pairing family

The pairing matrix is first specified in the orbital basis. We write

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta \left(M_0 + \eta M_1\right), \tag{5}$$

with

$$M_0 = \tau_0\sigma_0, \qquad M_1 = \tau_x\sigma_x. \tag{6}$$

Here τ_i act in layer space and σ_i act in sublattice space. The parameter η interpolates between the reference orbital-singlet direction and the interlayer-sublattice-mixed direction. The normalization $\mathcal{N}_\eta$ is treated as a control variable. We use two conventions:

$$\text{fixed Frobenius:} \qquad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \tag{7}$$

$$\text{fixed } \Delta_0: \qquad \mathcal{N}_\eta = 1. \tag{8}$$

At each momentum, the orbital pairing is projected into the retained band subspace as

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k})\Delta_{\text{orb}}U_-^*(-\mathbf{k}). \tag{9}$$

The working valley convention is a time-reversal-sewn proxy, $U_-(-\mathbf{k}) = U_+^*(\mathbf{k})$. Thus the production scans use $\Delta_{\text{band}} = U_+^\dagger\Delta_{\text{orb}}U_+$ after the valley basis is sewn. A raw independently diagonalized valley-minus basis is not used for W_{inter} because it is not automatically in this sewing convention. This is used as a controlled diagnostic convention rather than a claim that all valley-basis subtleties are solved. The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n\neq m}|\Delta_{nm}|^2}{\sum_{n,m}|\Delta_{nm}|^2}. \tag{10}$$

III. FINITE-BAND RESPONSE

For each retained band subspace we construct a finite-band BdG Hamiltonian,

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\text{band}}(\mathbf{k}) \\ \Delta_{\text{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \tag{11}$$

The static current-current response is evaluated at zero temperature as

$$\Lambda_{\alpha\beta} = \sum_{a\neq b} \frac{\langle a|J_\alpha|b\rangle\langle b|J_\beta|a\rangle(f_a - f_b)}{E_b - E_a}. \tag{12}$$

The values reported by the executable response engine are finite-band raw diagnostics. They are internally consistent response outputs, but they are not used here as final experimental stiffness values.

The band-basis velocity is decomposed into diagonal and off-diagonal parts,

$$v_\alpha^{\text{band}} = v_\alpha^{\text{intra}} + v_\alpha^{\text{inter}}. \tag{13}$$

This yields conventional, geometric, and cross response channels. The isotropic response used in the mechanism scans is

$$D_{\text{iso}} = \frac{1}{2}(D_{xx} + D_{yy}). \tag{14}$$

The numerical checks track BdG Hermiticity, particle-hole symmetry, and exact closure of the channel decomposition.

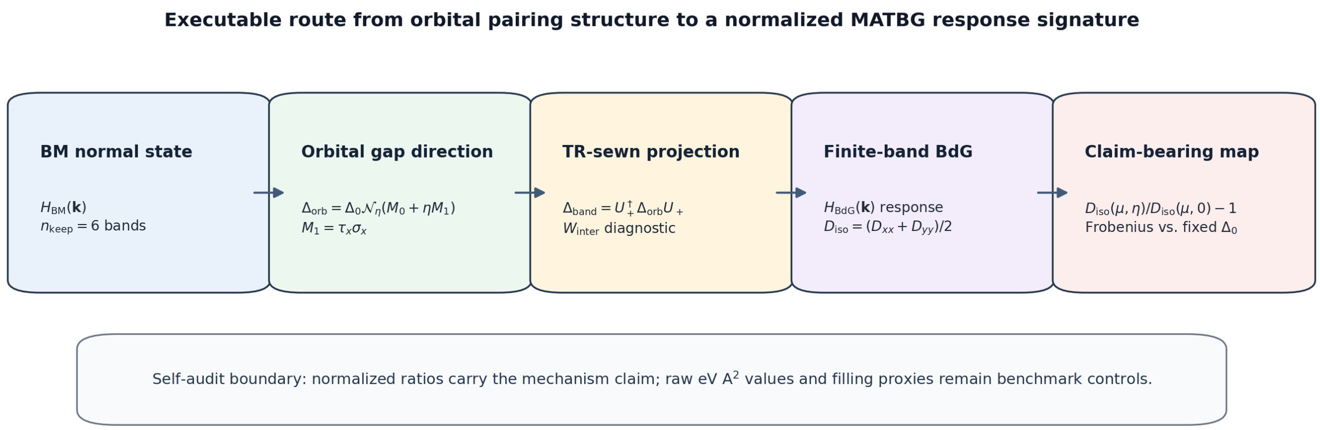

FIG. 1. Workflow used to convert an orbital-defined pairing direction into a normalized finite-band response diagnostic. The BM normal-state bands define the retained subspace, the orbital pairing matrix is projected with the time-reversal-sewn valley convention, the BdG response is evaluated in the retained band space, and the claim-bearing observable is the normalized response ratio. Absolute eV Å^2 values and filling proxies remain benchmark controls in the present mechanism-paper scope.

IV. PAIRING GATES AND RESPONSE MAPS

A. Pairing and velocity gates

The first algebraic pairing gate verifies that an orbital-defined pairing family transforms covariantly under band-basis gauge changes. The BM projection gate then selects $M_1 = \tau_x \sigma_x$ as a useful working direction because it produces a finite projected interband weight in the time-reversal-sewn valley convention while keeping $W_{\mathrm{inter}} = 0$ for $\eta = 0$. For $n_{\mathrm{keep}} = 6$, $nk = 7$, and $\eta = 1$, the dense scan gives

$$W_{\mathrm{inter}} \simeq 0.108. \tag{15}$$

The velocity gate confirms that projected velocity matrices are Hermitian to machine precision and that the intra/inter velocity split closes numerically. These gates are not yet response claims. They establish that the pairing direction, valley sewing convention, and current decomposition are internally consistent before the normalized response maps are interpreted.

B. Dense chemical-potential and pairing scan

Table I summarizes the dense scan at $nk = 7$ and $n_{\mathrm{keep}} = 6$. It is the main endpoint comparison: each row compares $\eta = 1$ with the corresponding $\eta = 0$ reference at the same chemical potential. The fixed-Frobenius column asks how the response changes when the orbital direction is changed at fixed matrix norm, while the fixed-Δ_0 column keeps the raw gap amplitude fixed and therefore serves as the normalization control. The first column gives the retained-band filling proxy ν_{proxy}; this is a diagnostic occupancy proxy and is not used as a calibrated experimental filling. The Supplemental Material gives a central two-flat-band crosswalk; in that convention the same chemical-potential window spans approximately $\nu_{\mathrm{flat}} = -4$ to $+4$. A filling-sufficiency audit verifies that this crosswalk is adequate as a BM central-flat-band counting reference for the present mechanism maps, but not as a device-level carrier-density calibration.

Figure 2 shows the fixed-Frobenius dense response map. The table records the $\eta = 1$ endpoint, while the heatmap shows how the response evolves throughout the interpolation. At $\eta = 1$ the fixed-Frobenius response is positive for every tabulated chemical potential, with the largest endpoint value in the present scan occurring near $\mu = 5$ meV. The middle panel shows that the geometric fraction remains a useful diagnostic of how the response is redistributed between current sectors.

Figure 3 shows the same scan at fixed raw Δ_0. The response is much weaker and can be negative. This control is essential: it shows that the fixed-Frobenius map should be interpreted as a response to a specified normalized orbital direction, not as a normalization-independent enhancement produced by any finite interband weight.

C. $nk = 9$ validation

The key-point validation in Table II repeats representative values at $nk = 9$. It is not a replacement for the dense $nk = 7$ map; rather, it tests whether the main sign and scale survive on a finer momentum grid at chemically distinct points. The fixed-Frobenius signal remains positive and of comparable size, while the fixed-Δ_0 response stays close to zero. Additional selected $nk = 11$ and $nk = 13$ key-point checks, reported in

TABLE I. Dense μ scan comparing $\eta = 1$ to $\eta = 0$. The fixed-Frobenius column reports $D_{\mathrm{iso}}(\eta = 1)/D_{\mathrm{iso}}(\eta = 0) - 1$. The fixed-$\Delta_0$ column reports the same quantity without rescaling the orbital pairing matrix.

μ (meV)	ν_{proxy}	fixed Frobenius	fixed Δ_0
-5	-1.483	0.0056	-0.0510
-4	-1.333	0.0891	0.0291
-3	-1.061	0.0785	-0.0071
-2	-0.857	0.0537	-0.0110
-1	-0.653	0.0802	0.0093
0	-0.381	0.0817	0.0112
1	-0.054	0.0914	0.0044
2	0.122	0.0949	0.0119
3	0.395	0.0653	0.0004
4	0.667	0.0662	-0.0005
5	0.857	0.1186	0.0312

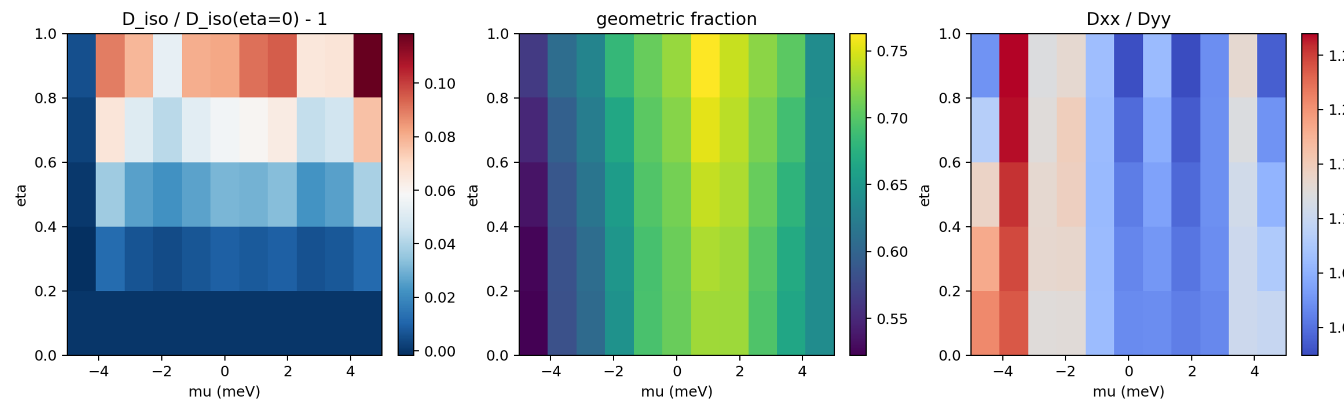

FIG. 2. Dense μ-η response map in the fixed-Frobenius normalization. Left: normalized isotropic response relative to $\eta = 0$ at fixed μ. Middle: geometric fraction. Right: response anisotropy D_{xx}/D_{yy}.

TABLE II. $nk = 9$, $n_{\mathrm{keep}} = 6$ validation of the $\eta = 1$ response.

μ (meV)	fixed Frobenius	fixed Δ_0
-4	0.0604	-0.0068
0	0.0863	0.0103
2	0.0643	-0.0029
4	0.0605	-0.0017

the Supplemental Material, preserve the positive fixed-Frobenius response at the same chemical potentials. A finite-grid trend audit of these key points versus $1/nk^2$ gives positive fixed-Frobenius intercepts between 8.65% and 10.58%, while the fixed-Δ_0 control remains weak and sign-sensitive on the measured grids. The Supplemental Material also reports a targeted $nk = 15$ spot check at $\mu = 0$ and 2 meV, where the fixed-Frobenius response remains +8.97% and +8.65%, respectively. A convergence-sufficiency self-audit combines these selected-grid, trend, and spot-check data. All audit gates pass for the selected-grid normalized mechanism claim, while the same audit explicitly does not promote the numbers to a strict continuum-limit value.

V. SELF-AUDIT OF ABSOLUTE UNITS

The raw response engine outputs values in finite-band diagnostic units. A conservative conversion maps raw meV Å^2 to eV Å^2 per flavor by dividing by 1000. Any spin/valley degeneracy factor is left outside this conversion:

$$D_{\text{per flavor}}[\mathrm{eV\,\AA}^2] = D_{\mathrm{raw}}[\mathrm{meV\,\AA}^2]/1000. \tag{16}$$

The moire area and reciprocal scale agree with the standard MATBG geometry, but the prior band-basis decomposition benchmark table [23] is not reproduced by a single global prefactor or by the present production convention. The claim-bearing observable policy for the present manuscript is therefore: the production τ_z/τ_0 finite-band response convention is used for normalized ratios and channel diagnostics, while eV Å^2 per-flavor values are restricted to benchmark provenance and self-audit tables.

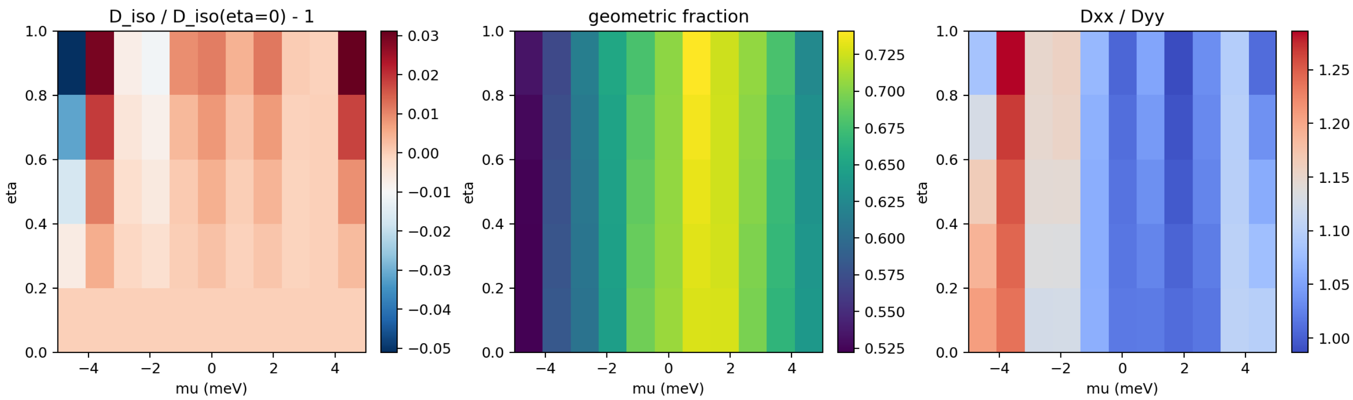

FIG. 3. Dense μ-η response map in the fixed-Δ_0 normalization. Compared with Fig. 2, the normalized response is strongly suppressed and changes sign in parts of the scan.

Table III summarizes the reconstruction audit for uniform s-wave pairing at $\mu = 0$, $\Delta_0 = 1$ meV, and $nk = 14$. The best simple unified reconstruction is

$$D_{\mathrm{conv}} = 2D_{\mathrm{intra},\tau_z}, \qquad D_{\mathrm{geom}} = D_{\mathrm{inter},\tau_z}. \tag{17}$$

It nearly reconstructs the extended-band endpoint but not the flat-band endpoint. A separate flat-band audit shows that the old $n_{\mathrm{keep}} = 2$ total and conventional values can be approximated with a Gamma-centered mesh and a full curvature-like conventional correction, but that correction does not work cleanly at $n_{\mathrm{keep}} = 6$.

This self-audit changes the interpretation of absolute stiffness values. The historical table is useful as a benchmark, but it should not be used to set the convention for the present interband-pairing mechanism scans. If a later version adds direct experimental comparison, absolute tables should be regenerated from one declared response convention, with degeneracy, Brillouin-zone normalization, and any diamagnetic or curvature term stated explicitly.

VI. DISCUSSION

The main physical message is that an interband-pairing direction can leave a detectable imprint on the normalized finite-band response of MATBG. The imprint is reproducible across chemical potential and survives a modest increase of the momentum grid, but it depends strongly on how the orbital gap direction is normalized. This is both a result and a warning: interband pairing is not a scalar knob. Changing the orbital structure can also change the total norm of the projected gap, and the resulting response should not be interpreted without specifying the normalization convention.

The self-audit also clarifies the status of absolute stiffness claims. The continuum BM model has subtleties associated with UV regularization, current convention, and band truncation. Those subtleties are manageable, but the audit shows that old absolute tables can mix conventions in ways that are not captured by one simple prefactor. For this reason the most robust near-term observable in the present work is a normalized response map, not an absolute stiffness prediction. Likewise, ν_{proxy} in the tables is a retained-band occupancy proxy defined for scan organization, not a calibrated carrier density. The Supplemental Material gives a central two-flat-band filling crosswalk for the dense scan, but the present figures should still be read as functions of chemical potential and retained-band occupancy, not as a quantitative map onto an experimental superconducting dome. Within that restricted use, the filling-sufficiency audit shows that the dense scan spans the full central two-flat-band counting range and samples reference points near $\nu_{\mathrm{flat}} = 0$ and ± 2. The Supplemental Material also includes a claim-scope audit that separates the allowed normalized-response mechanism claims from deferred absolute-stiffness, device-calibration, and continuum-limit claims.

The evidence chain is therefore intentionally narrower than a full microscopic theory of MATBG superconductivity. The pairing gates establish that the orbital construction creates a finite band-off-diagonal component after projection; the dense maps show that this component changes the normalized response when the orbital pairing direction is compared at fixed Frobenius norm; the fixed-Δ_0 maps show that the same statement is not normalization independent; and the selected-grid audits test that the sign and scale are not artifacts of the smallest momentum grid. The interpretation would be weakened if the projected interband weight became unstable under the retained-band or grid choices, if the fixed-Frobenius signal lost its sign stability on the selected higher grids, or if the fixed-Δ_0 control showed the same response pattern. None of these checks replaces a self-consistent pairing calculation, but they make the present conclusion falsifiable within the finite-band diagnostic framework used here.

Several limitations remain. First, the valley sewing convention used here is a controlled time-reversal proxy; a fully independent two-valley implementation would

TABLE III. Prior band-basis benchmark reconstruction. Values are eV Å^2 per flavor. The reconstruction route is kept separate from the production interband-pairing scans.

case	n_{keep}	D_{total}	target	D_{conv}	target	D_{geom}
current τ_z/τ_0	2	29.34	67.50	20.95	53.00	8.40
double-conv all-τ_z	2	54.40	67.50	41.89	53.00	12.50
flat endpoint audit	2	66.18	67.50	53.83	53.00	12.36
double-conv all-τ_z	6	126.66	129.30	51.45	54.00	75.22

be required for valley-asymmetric perturbations or final absolute production calculations. Second, a direct experimental-comparison version would require a final absolute response convention; the present mechanism paper therefore makes no claim-bearing absolute stiffness prediction. Third, the filling variable is currently a retained-band occupancy proxy. Fourth, no self-consistent gap equation is solved. Fifth, a UV-complete lattice calculation would be required for final quantitative comparison with experiment. Recent direct imaging of interaction-reshaped flat bands in MATBG also suggests that a future quantitative comparison should revisit the normal-state band structure beyond the noninteracting BM spectrum used here [24]. Finally, the present ansatz is deliberately simpler than recent microscopic band-off-diagonal and finite-momentum scenarios. Recent tunneling and STM work in magic-angle graphene reports nodal-gap, separated-gap, and intervalley features, while theory has connected band-off-diagonal and Kekulé pair-density-wave structures to spectral and stiffness signatures [20, 21, 25–28]. Those states are complementary to the present uniform orbital-projected diagnostic and will require a separate finite-momentum BdG extension.

VII. CONCLUSIONS

The calculation isolates a concrete finite-band diagnostic of pairing structure in MATBG. For the orbital direction $M_1 = \tau_x\sigma_x$, the projected interband weight is finite and the fixed-Frobenius response map shows a reproducible positive change across the scanned chemical-potential window. The fixed-Δ_0 control, by contrast, is weak and sign-sensitive. This contrast is the central result: interband pairing produces a normalization-conditioned mechanism signature in the finite-band MATBG superfluid response, not an absolute stiffness prediction independent of how the gap direction is normalized.

The self-audit of units and filling fixes the scope of this conclusion. Normalized response ratios are robust claim-bearing observables for the present mechanism paper, while absolute stiffness tables, device-level filling calibration, and continuum-limit extrapolation remain separate tasks. Within that boundary, the work provides an executable route for comparing orbital gap structure, band projection, and quantum-geometric response in magic-angle graphene superconductivity, together with explicit checks that can invalidate or sharpen the mechanism claim in future higher-fidelity calculations.

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[1] Y. Cao, V. Fatemi, S. Fang, K. Watanabe, T. Taniguchi, E. Kaxiras, and P. Jarillo-Herrero, Unconventional superconductivity in magic-angle graphene superlattices, Nature **556**, 43 (2018).

[2] Y. Cao, V. Fatemi, A. Demir, S. Fang, S. L. Tomarken, J. Y. Luo, J. D. Sanchez-Yamagishi, K. Watanabe, T. Taniguchi, E. Kaxiras, R. C. Ashoori, and P. Jarillo-Herrero, Correlated insulator behaviour at half-filling in magic-angle graphene superlattices, Nature **556**, 80 (2018).

[3] M. Yankowitz, S. Chen, H. Polshyn, Y. Zhang, K. Watanabe, T. Taniguchi, D. Graf, A. F. Young, and C. R. Dean, Tuning superconductivity in twisted bilayer graphene, Science **363**, 1059 (2019).

[4] S. Peotta and P. Törmä, Superfluidity in topologically nontrivial flat bands, Nature Communications **6**, 8944 (2015).

[5] P. Törmä, S. Peotta, and B. A. Bernevig, Superconductivity, superfluidity and quantum geometry in twisted multilayer systems, Nature Reviews Physics **4**, 528

(2022).

[6] Z. Song, Z. Wang, W. Shi, G. Li, C. Fang, and B. A. Bernevig, All magic angles in twisted bilayer graphene are topological, Physical Review Letters **123**, 036401 (2019).

[7] H. C. Po, L. Zou, T. Senthil, and A. Vishwanath, Faithful tight-binding models and fragile topology of magic-angle bilayer graphene, Physical Review B **99**, 195455 (2019).

[8] F. Xie, Z. Song, B. Lian, and B. A. Bernevig, Topology-bounded superfluid weight in twisted bilayer graphene, Physical Review Letters **124**, 167002 (2020).

[9] H. Tian, X. Gao, Y. Zhang, *et al.*, Evidence for Dirac flat band superconductivity enabled by quantum geometry, Nature **614**, 440 (2023).

[10] M. Tanaka, J. I.-J. Wang, T. H. Dinh, *et al.*, Superfluid stiffness of magic-angle twisted bilayer graphene, Nature **638**, 99 (2025).

[11] E. Portolés, M. Perego, P. A. Volkov, *et al.*, Quasiparticle and superfluid dynamics in magic-angle graphene, Nature Communications **16**, 4273 (2025).

[12] A. Banerjee, Z. Hao, M. Kreidel, *et al.*, Superfluid stiffness of twisted trilayer graphene superconductors, Nature **638**, 93 (2025).

[13] X. Hu, T. Hyart, D. I. Pikulin, and E. Rossi, Geometric and conventional contribution to the superfluid weight in twisted bilayer graphene, Physical Review Letters **123**, 237002 (2019).

[14] A. Julku, T. J. Peltonen, L. Liang, T. T. Heikkilä, and P. Törmä, Superfluid weight and Berezinskii-Kosterlitz-Thouless transition temperature of twisted bilayer graphene, Physical Review B **101**, 060505 (2020).

[15] N. Verma, T. Hazra, and M. Randeria, Optical spectral weight, phase stiffness, and T_c bounds for trivial and topological flat band superconductors, Proceedings of the National Academy of Sciences **118**, e2106744118 (2021).

[16] C. Chen, K. P. Nuckolls, S. Ding, *et al.*, Strong electron–phonon coupling in magic-angle twisted bilayer graphene, Nature **636**, 342 (2024).

[17] X. Gao, A. Jimeno-Pozo, P. A. Pantaleon, *et al.*, Double-edged role of interactions in superconducting twisted bilayer graphene, Nature Physics **22**, 692 (2026).

[18] Y.-J. Wang, G.-D. Zhou, S.-Y. Peng, B. Lian, and Z.-D. Song, Molecular pairing in twisted bilayer graphene superconductivity, Physical Review Letters **133**, 146001 (2024).

[19] M. S. Liang, Y.-J. Wang, G.-D. Zhou, Z.-D. Song, and X. Dai, Strongly correlated superconductivity in twisted bilayer graphene: A gutzwiller study (2026), arXiv:2604.04631 [cond-mat.str-el].

[20] M. Christos, S. Sachdev, and M. S. Scheurer, Nodal band-off-diagonal superconductivity in twisted graphene superlattices, Nature Communications **14**, 7134 (2023).

[21] B. Putzer and M. S. Scheurer, Eliashberg theory and superfluid stiffness of band-off-diagonal pairing in twisted graphene, Physical Review B **111**, 144513 (2025).

[22] R. Bistritzer and A. H. MacDonald, Moiré bands in twisted double-layer graphene, Proceedings of the National Academy of Sciences **108**, 12233 (2011).

[23] J. Zhou, Band-basis decomposition of superfluid weight in magic-angle twisted bilayer graphene: Quantifying geometric and conventional contributions (2026), arXiv:2604.05994 [cond-mat.supr-con].

[24] J. Xiao, A. Inbar, J. Birkbeck, N. Gershon, Y. Zamir, Y. Vituri, T. Taniguchi, K. Watanabe, E. Berg, and S. Ilani, Imaging the flat bands of magic-angle graphene reshaped by interactions, Nature **653**, 68 (2026).

[25] J. M. Park, S. Sun, K. Watanabe, T. Taniguchi, and P. Jarillo-Herrero, Experimental evidence for nodal superconducting gap in moiré graphene, Science **391**, 79 (2025).

[26] H. Kim, G. Rai, L. Crippa, *et al.*, Resolving intervalley gaps and many-body resonances in moiré superconductors, Nature **650**, 592 (2026).

[27] K. Wang and K. Levin, Kekulé superconductivity in twisted magic angle bilayer graphene (2026), arXiv:2510.06451 [cond-mat.supr-con].

[28] K. Wang, Q. Chen, R. Boyack, and K. Levin, Geometric superfluid stiffness of Kekulé superconductivity in magic-angle twisted bilayer graphene (2026), arXiv:2603.24766 [cond-mat.supr-con].